<table>
<tr><th>Q n</th><th>Solutions</th><th>Comments</th></tr>
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$e^{2x}\ln(1+ax)$

$=\left(1+2x+\frac{(2x)^2}{2!}+\dots\right)\left(ax-\frac{(ax)^2}{2}+\frac{(ax)^3}{3}-\dots\right)$

$=ax-\frac{a^2x^2}{2}+\frac{a^3x^3}{3}+2ax^2-a^2x^3+2ax^3+\dots$

$=ax+\left(2a-\frac{a^2}{2}\right)x^2+\left(\frac{a^3}{3}-a^2+2a\right)x^3+\dots$

$2a-\frac{a^2}{2}=0$

$a\left(2-\frac{a}{2}\right)=0$

$a=4$ or $0(NA)$

$\therefore a=4$

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For no term in x^2, it means the coefficient of x^2 is 0.

Read question carefully, it is mentioned that a is non-zero.

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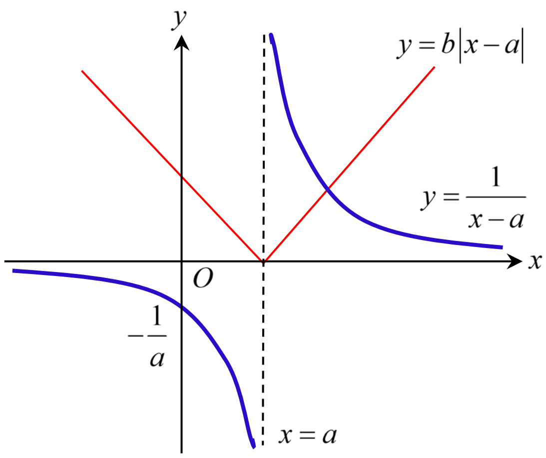
<tr><td>2i</td><td>



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Although there are unknowns a and b, we can have a "sense" of the graphs in GC by setting the values of a and b to be positive numbers.

The point of intersection P is used for the next part in the question.

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<tr><td>ii</td><td>

At point P, $b(x-a)=\frac{1}{x-a}$

$(x-a)^2=\frac{1}{b}$

$x-a=\frac{1}{\sqrt{b}}$ or $-\frac{1}{\sqrt{b}}(NA)$

$x=a+\frac{1}{\sqrt{b}}$

From the graph, $\frac{1}{x-a}<b|x-a|$

$x<a$ or $x>a+\frac{1}{\sqrt{b}}$.

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We drop the modulus because from the graph in (i), the x-coordinate of P is at $x>a$.

From the argument above, $x-a>0$.

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3i	$y^2-2xy+5x^2-10=0$ $2y\frac{\mathrm{d}y}{\mathrm{d}x}-2y-2x\frac{\mathrm{d}y}{\mathrm{d}x}+10x=0$ At stationary points, $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ $-2y + 10x = 0$ $y = 5x$ Substitute $y = 5x$ into the equation $(5x)^2-2x(5x)+5x^2-10=0$ $20x^2=10$ $x=\pm\frac{1}{\sqrt{2}}$ The exact x-coordinates of the stationary points of C are $\frac{1}{\sqrt{2}}$ and $-\frac{1}{\sqrt{2}}$.	There is no need to spend time making $\frac{\mathrm{d}y}{\mathrm{d}x}$ the subject. We choose to express y in terms of x because our aim is to find values of x, so we want to substitute y. Read question carefully, we need the answers in exact form. Do NOT use GC like stated in the question.
	$2y\frac{\mathrm{d}y}{\mathrm{d}x}-2y-2x\frac{\mathrm{d}y}{\mathrm{d}x}+10x=0$ $2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2+2y\frac{\mathrm{d}^2y}{\mathrm{d}x^2}-2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)-2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)-2x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+10=0$ When $\frac{\mathrm{d}y}{\mathrm{d}x}=0$, $2y\frac{\mathrm{d}^2y}{\mathrm{d}x^2}-2x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+10=0$ When $x>0, x=\frac{1}{\sqrt{2}}, y=\frac{5}{\sqrt{2}}$ $2\left(\frac{5}{\sqrt{2}}\right)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}-2\left(\frac{1}{\sqrt{2}}\right)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+10=0$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=-\frac{5\sqrt{2}}{4}<0$ The stationary point is a maximum.	Always differentiate before substituting values. It was a stationary point Substitute the values of x and y, when $x > 0$.
4i	$y=\frac{4x+9}{x+2}$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{4(x+2)-(4x+9)}{(x+2)^2}$ $=-\frac{1}{(x+2)^2}$ Since $(x + 2)^2 > 0$ for all real x, $x\neq-2$, gradient of C is negative for all points on C.	Usually, we say that $(x+2)^2\geq0$. However, since $x + 2$ came from the denominator, $x\neq2$. Hence, $(x+2)^2\neq0$.

ii	$y=\dfrac{4x+9}{x+2}=4+\dfrac{1}{x+2}$, $a=4$ and $b=1$ Asymptotes are $y = 4$ and $x=-2$.	
iii	$\left(y=4+\dfrac{1}{x+2}\rightarrow y=4+\dfrac{1}{(x-2)+2}=4+\dfrac{1}{x}\right)$ Translate the graph of C 2 units in the positive x-direction. $\left(y-4=\dfrac{1}{x}\rightarrow (y+4)-4=\dfrac{1}{x}\right)$ Translate the resultant graph 4 units in the negative y-direction.	The explanations in the brackets are working for us to track the equation after every transformation. They are NOT the transformations. The order of the transformations does not matter for this question.
5i	Let $f(x) = x^3 + ax^2 + bx + c$ $f(1) = 1 + a + b + c = 8$ $a + b + c = 7$ $f(2) = 8 + 4a + 2b + c = 12$ $4a + 2b + c = 4$ $f(3) = 27 + 9a + 3b + c = 25$ $9a + 3b + c = -2$ By GC, $a = -1.5$, $b = 1.5$, $c = 7$	This is an application of the remainder theorem learnt in secondary school.
ii	$y = x^3 - 1.5x^2 + 1.5x + 7$ $\dfrac{dy}{dx}=3x^2-3x+1.5$ <u>Method 1:</u> $\dfrac{dy}{dx}=3\left(\left(x-\dfrac{1}{2}\right)^2-\dfrac{1}{4}\right)+1.5=3\left(x-\dfrac{1}{2}\right)^2+0.75$ Since $\left(x-\dfrac{1}{2}\right)^2\geq 0$, $\dfrac{dy}{dx}=3\left(x-\dfrac{1}{2}\right)^2+0.75\geq 0.75>0$ Hence the gradient of C is always positive. <u>Method 2:</u> Since $(-3)^2-4(3)(1.5)=-9<0$ and the coefficient of $x^2 > 0$, $\dfrac{dy}{dx}>0$. Hence the gradient of C is always positive. Since gradient of C is always positive, the curve $y=f(x)$ is increasing for all real x. Furthermore, $f(-2)=-10<0$ and $f(0)=7>0$. Hence, the curve will cut the x-axis only once and $f(x)=0$ has only one solution. By GC, $x=-1.33$.	$b^2-4ac<0$ only means there is no real root, i.e. it could be always positive or always negative. This is a hence question, need to use the fact that gradient is always positive.
iii	$\dfrac{dy}{dx}=3x^2-3x+1.5=2$ $3x^2-3x-0.5=0$ By GC, $x=1.1455$ or -0.14549 The x-coordinates are 1.15 and -0.145.	

6i	$\underset{\sim}{r}=\underset{\sim}{a}+t\underset{\sim}{b}$ is the vector equation of a straight line passing through a point A with position vector $\underset{\sim}{a}$ and the line is parallel to the vector $\underset{\sim}{b}$.	The line does NOT pass through $\underset{\sim}{a}$. The line passes through a point A with position vector $\underset{\sim}{a}$.
ii	$\underset{\sim}{r}\cdot\underset{\sim}{n}=d$ is the vector equation of a plane with normal vector parallel to $\underset{\sim}{n}$ and perpendicular distance $\lvert d\rvert$ units from origin to the plane. So the numerical value of d or $\lvert d\rvert$ is the shortest distance from the origin to the plane.	The significance of d in this question works because $\underset{\sim}{n}$ is a unit vector. Refer to Vectors tutorial 6
iii	$\left(\underset{\sim}{a}+t\underset{\sim}{b}\right)\cdot\underset{\sim}{n}=d$ $\underset{\sim}{a}\cdot\underset{\sim}{n}+t\underset{\sim}{b}\cdot\underset{\sim}{n}=d$ $t=\dfrac{d-\underset{\sim}{a}\cdot\underset{\sim}{n}}{\underset{\sim}{b}\cdot\underset{\sim}{n}}\left(\text{since } \underset{\sim}{b}\cdot\underset{\sim}{n}\neq 0\right)$ $\therefore \underset{\sim}{r}=\underset{\sim}{a}+\left(\dfrac{d-\underset{\sim}{a}\cdot\underset{\sim}{n}}{\underset{\sim}{b}\cdot\underset{\sim}{n}}\right)\underset{\sim}{b}$ Since $\underset{\sim}{b}\cdot\underset{\sim}{n}\neq 0$, the line is not parallel to the plane. Hence, they will intersect at a point given by the position vector $\underset{\sim}{a}+\left(\dfrac{d-\underset{\sim}{a}\cdot\underset{\sim}{n}}{\underset{\sim}{b}\cdot\underset{\sim}{n}}\right)\underset{\sim}{b}$.	This part is about finding the point of intersection between a line and a plane, just without numbers. Similar to 6i, $\underset{\sim}{a}+\left(\dfrac{d-\underset{\sim}{a}\cdot\underset{\sim}{n}}{\underset{\sim}{b}\cdot\underset{\sim}{n}}\right)\underset{\sim}{b}$ is not a point, it is position vector of the point.
7i	$\int \sin 2mx \sin 2nx \,\mathrm{d}x$ $=-\dfrac{1}{2}\int\left[\cos 2(m+n)x-\cos 2(m-n)x\right]\,\mathrm{d}x$ $=-\dfrac{1}{2}\left[\dfrac{\sin 2(m+n)x}{2(m+n)}-\dfrac{\sin 2(m-n)x}{2(m-n)}\right]+C$ $=\dfrac{\sin 2(m-n)x}{4(m-n)}-\dfrac{\sin 2(m+n)x}{4(m+n)}+C$	
ii	$\left(\mathrm{f}(x)\right)^2=\sin^2 2mx+\sin^2 2nx+2\sin 2mx\sin 2nx$ $=\dfrac{1-\cos 4mx}{2}+\dfrac{1-\cos 4nx}{2}+2\sin 2mx\sin 2nx$ $=1-\dfrac{\cos 4mx}{2}-\dfrac{\cos 4nx}{2}+2\sin 2mx\sin 2nx$ $\int_0^{\pi}\left(\mathrm{f}(x)\right)^2\,\mathrm{d}x$ $=\int_0^{\pi}1-\dfrac{\cos 4mx}{2}-\dfrac{\cos 4nx}{2}+2\sin 2mx\sin 2nx\,\mathrm{d}x$ $=\left[x-\dfrac{\sin 4mx}{8m}-\dfrac{\sin 4nx}{8n}+2\left(\dfrac{\sin 2(m-n)x}{4(m-n)}-\dfrac{\sin 2(m+n)x}{4(m+n)}\right)\right]_0^{\pi}$ $=\pi$	Use result from **(i)**. $\sin k\pi=0$ for integer k.
8a	$z^2\left(1-\mathrm{i}\right)-2z+\left(5+5\mathrm{i}\right)=0$	For this question, please read the instructions. We

	$z=\dfrac{-(-2)\pm\sqrt{(-2)^2-4(1-\mathrm{i})(5+5\mathrm{i})}}{2(1-\mathrm{i})}$ $=\dfrac{2\pm\sqrt{4-20(1-\mathrm{i})(1+\mathrm{i})}}{2(1-\mathrm{i})}$ $=\dfrac{2\pm\sqrt{4-20(2)}}{2(1-\mathrm{i})}$ $=\dfrac{2\pm\sqrt{-36}}{2(1-\mathrm{i})}$ $=\dfrac{2\pm 6\mathrm{i}}{2(1-\mathrm{i})}$ $=\dfrac{1+3\mathrm{i}}{1-\mathrm{i}}$ or $\dfrac{1-3\mathrm{i}}{1-\mathrm{i}}$ $=\dfrac{(1+3\mathrm{i})(1+\mathrm{i})}{(1-\mathrm{i})(1+\mathrm{i})}$ or $\dfrac{(1-3\mathrm{i})(1+\mathrm{i})}{(1-\mathrm{i})(1+\mathrm{i})}$ $=\dfrac{1+3\mathrm{i}+\mathrm{i}-3}{2}$ or $\dfrac{1-3\mathrm{i}+\mathrm{i}+3}{2}$ $=-1+2\mathrm{i}$ or $2-\mathrm{i}$	are not allowed to use calculators in this question, so in the process of solving, we need to show working clearly, especially for evaluating the discriminant, division of complex number etc. While we are not allowed to use the calculators in the solving, we can use calculators to check our answers.
bi	$w=1-\mathrm{i}$ $w^2=(1-\mathrm{i})^2=1-2\mathrm{i}+(-1)=-2\mathrm{i}$ $w^3=-2\mathrm{i}(1-\mathrm{i})=-2-2\mathrm{i}$ $\mathrm{w}^4=(-2\mathrm{i})^2=-4$ Subst $w=1-\mathrm{i}$ into $w^4+pw^3+39w^2+qw+58=0$ $-4+p(-2-2\mathrm{i})+39(-2\mathrm{i})+q(1-\mathrm{i})+58=0$ $(-4-2p+q+58)+\mathrm{i}(-2p-78-q)=0$ Comparing real and imaginary parts, $-2p+q+54=0\Rightarrow q=2p-54$ $-2p-q-78=0\Rightarrow q=-2p-78$ $2p-54=-2p-78$ $4p=-24$ $p=-6,\quad q=-66$	Same comment in use of calculator for this part. We could also solve this in the polar form, however, repeated converting between forms could be a hassle. Since we are not to use calculators for this question, we need to show how we solve the simultaneous equations. We can use calculators to check our answers.
ii	Since all the coefficients of $w^4-6w^3+39w^2-66w+58=0$ are real, $w=1+\mathrm{i}$ is also a root to the equation. $(w-(1-\mathrm{i}))(w-(1+\mathrm{i}))=w^2-2w+2$ <u>Method 1</u>	We will make use of the fact that factors from conjugate roots will give us a quadratic factor easily. $w=1-\mathrm{i}$ is a root to the equation, $(w-(1-\mathrm{i}))$ is a factor to the expression.

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$$
\begin{array}{r|rrrrrrrrr}
 & & & w^2 & - & 4w & + & 29 & & \\
\hline
w^2-2w+2 & w^4 & - & 6w^3 & + & 39w^2 & - & 66w & + & 58 \\
- & (w^4 & - & 2w^3 & + & 2w^2) & & & & \\
\hline
 & & - & 4w^3 & + & 37w^2 & - & 66w & + & 58 \\
 & - & (- & 4w^3 & + & 8w^2 & - & 8w) & & \\
\hline
 & & & & & 29w^2 & - & 58w & + & 58 \\
 & & & & - & (29w^2 & - & 58w & + & 58) \\
\hline
 & & & & & & & & & 0
\end{array}
$$

The quadratic factors are w^2-2w+2 and $w^2-4w+29$.

<u>Method 2</u>
$w^4-6w^3+39w^2-66w+58=\left(w^2-2w+2\right)\left(aw^2+bw+c\right)$
Compare coefficient of w^4: $a=1$
Compare constant term: $2c=58\Rightarrow c=29$
Compare coefficient of w^3: $b-2=-6\Rightarrow b=-4$

The quadratic factors are w^2-2w+2 and $w^2-4w+29$.
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<td>We can actually write down the values of a and c by observation, (i.e. without working).</td>
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$$
\begin{aligned}
u_n &= S_n - S_{n-1} \\
&= An^2+Bn-\left(A(n-1)^2+B(n-1)\right) \\
&= An^2+Bn-An^2+2An-A-Bn+B \\
&= 2An-A+B \\
&= A(2n-1)+B
\end{aligned}
$$
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<tr>
<td>ii</td>
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$u_{10}=19A+B=48$
$u_{17}=33A+B=90$
By GC, $A=3$, $B=-9$
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<td>b</td>
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$$
\begin{aligned}
r^2(r+1)^2-(r-1)^2r^2 &= r^2\left(r^2+2r+1\right)-r^2\left(r^2-2r+1\right) \\
&= r^4+2r^3+r^2-r^4+2r^3-r^2 \\
&= 4r^3
\end{aligned}
$$

$$
\sum_{r=1}^{n}4r^3=\sum_{r=1}^{n}\left(r^2(r+1)^2-(r-1)^2r^2\right)
$$

$$
\begin{array}{rrcr}
 & 1^2 2^2 & - & 0^2 1^2 \\
 + & 2^2 3^2 & - & 1^2 2^2 \\
 = & & \vdots & \\
 + & (n-1)^2 n^2 & - & (n-2)^2(n-1)^2 \\
 + & n^2(n+1)^2 & - & (n-1)^2 n^2
\end{array}
$$

$$
=n^2(n+1)^2
$$

$$
\sum_{r=1}^{n}r^3=\frac{n^2(n+1)^2}{4}
$$
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<td>$r^2(r+1)^2-(r-1)^2r^2$ is a strong hint that we are going to do MOD in subsequent parts.</td>
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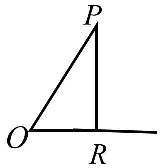
c	Let $a_r = \frac{x^r}{r!}$. $\left\|\frac{a_{r+1}}{a_r}\right\| = \left\|\frac{\frac{x^{r+1}}{(r+1)!}}{\frac{x^r}{r!}}\right\| = \frac{\|x\|}{r+1}$ For all real values of x, $\lim_{r\to\infty}\left\|\frac{a_{r+1}}{a_r}\right\| = \lim_{r\to\infty}\frac{\|x\|}{r+1} = 0 < 1$. By D'Alembert's ratio test, $\sum_{r=0}^{\infty}\frac{x^r}{r!}$ converges for all real values of x. $\sum_{r=0}^{\infty}\frac{x^r}{r!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^n}{n!} + \ldots$ $= \mathrm{e}^x$	Question gives us new information about test for $\sum_{r=0}^{\infty} a_r$ converging. Then we are required to test if $\sum_{r=0}^{\infty}\frac{x^r}{r!}$ converges, hence let $a_r = \frac{x^r}{r!}$. The rest of the proof follows the test where we need to find $\left\|\frac{a_{r+1}}{a_r}\right\|$, and determine if it is larger than 1, as $n \to \infty$. For any <u>fixed</u> value of x, $\lim_{r\to\infty}\frac{\|x\|}{r+1} = 0$. Last part of the question follows from the formula list. The polynomial series is a hint that we could be looking at a Maclaurin expansion.
10i	Equation of cable C is $\underset{\sim}{r} = s\begin{pmatrix}3\\1\\-2\end{pmatrix}, s \in \mathbb{R}$. $\overrightarrow{PQ} = \begin{pmatrix}5\\7\\a\end{pmatrix} - \begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}4\\5\\a+1\end{pmatrix}$ Equation of new cable is $\underset{\sim}{r} = \begin{pmatrix}1\\2\\-1\end{pmatrix} + t\begin{pmatrix}4\\5\\a+1\end{pmatrix}, t \in \mathbb{R}$. Since the cables intersect, $s\begin{pmatrix}3\\1\\-2\end{pmatrix} = \begin{pmatrix}1\\2\\-1\end{pmatrix} + t\begin{pmatrix}4\\5\\a+1\end{pmatrix}$ $3s \quad -4t \qquad = \quad 1 --(1)$ $s \quad -5t \qquad = \quad 2 --(2)$ $-2s \quad -t \quad -at \quad = \quad -1 --(3)$ By GC, $s = -\frac{3}{11}, \quad t = -\frac{5}{11}, \quad at = 2$ $a = \frac{at}{t} = \frac{2}{-\frac{5}{11}} = -\frac{22}{5}$	Both cables can be modelled by lines, so after removing context, this question is about intersection of 2 lines. Alternatively, we can find the values of s and t from equations (1) and (2), then find the values of t from (3).

ii	$\overrightarrow{OR}=s\begin{pmatrix}3\\1\\-2\end{pmatrix}$, for some $s\in\mathbb{R}$ $\overrightarrow{PR}=s\begin{pmatrix}3\\1\\-2\end{pmatrix}-\begin{pmatrix}1\\2\\-1\end{pmatrix}=\begin{pmatrix}3s-1\\s-2\\-2s+1\end{pmatrix}$ $\overrightarrow{QR}=s\begin{pmatrix}3\\1\\-2\end{pmatrix}-\begin{pmatrix}5\\7\\-3\end{pmatrix}=\begin{pmatrix}3s-5\\s-7\\-2s+3\end{pmatrix}$ Let θ be $\measuredangle PRQ$. $\cos\theta=\dfrac{\overrightarrow{PR}\cdot\overrightarrow{QR}}{\lvert\overrightarrow{PR}\rvert\lvert\overrightarrow{QR}\rvert}$ $=\dfrac{1}{\lvert\overrightarrow{PR}\rvert\lvert\overrightarrow{QR}\rvert}\left(\begin{pmatrix}3s-1\\s-2\\-2s+1\end{pmatrix}\cdot\begin{pmatrix}3s-5\\s-7\\-2s+3\end{pmatrix}\right)$ $=\dfrac{9s^2-18s+5+s^2-9s+14+4s^2-8s+3}{\lvert\overrightarrow{PR}\rvert\lvert\overrightarrow{QR}\rvert}$ $=\dfrac{14s^2-35s+22}{\lvert\overrightarrow{PR}\rvert\lvert\overrightarrow{QR}\rvert}$ Since $(-35)^2-4(14)(22)=-7<0$, $\cos\theta\neq 0$ for all real values of s, it is not possible for $\measuredangle PRQ$ to be $90°$.	Recall that in order to find $\measuredangle PRQ$, we need both vectors either converge to R or diverge from R. For this question, we want to check whether $\dfrac{\overrightarrow{PR}\cdot\overrightarrow{QR}}{\lvert\overrightarrow{PR}\rvert\lvert\overrightarrow{QR}\rvert}=0$, i.e. we are interested in the numerator. We do not need to evaluate $\lvert\overrightarrow{PR}\rvert\lvert\overrightarrow{QR}\rvert$ at all. Showing that the discriminant < 0 is enough to show that there is no solution. Contrast with Q4(i).
iii	$\overrightarrow{OR}=\left(\begin{pmatrix}1\\2\\-1\end{pmatrix}\cdot\dfrac{1}{\sqrt{14}}\begin{pmatrix}3\\1\\-2\end{pmatrix}\right)\dfrac{1}{\sqrt{14}}\begin{pmatrix}3\\1\\-2\end{pmatrix}$ $=\dfrac{7}{14}\begin{pmatrix}3\\1\\-2\end{pmatrix}$ $=\dfrac{1}{2}\begin{pmatrix}3\\1\\-2\end{pmatrix}$  The coordinates of R are $\left(\dfrac{3}{2},\dfrac{1}{2},-1\right)$.	If we remove the context, we are trying to find the foot of perpendicular from P to the line represented by cable C. A diagram is useful to help us visualize the context. An alternative is to solve this as a minimum problem, considering $\lvert\overrightarrow{PR}\rvert=\left\lvert\begin{pmatrix}3s-1\\s-2\\-2s+1\end{pmatrix}\right\rvert$ Answer in coordinates. Second part is equivalent to finding PR since we have found R from the previous part.

	$\overrightarrow{PR} = \begin{pmatrix} 1.5 \\ 0.5 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ $= \begin{pmatrix} 0.5 \\ -1.5 \\ 0 \end{pmatrix}$ $PR = \sqrt{(0.5)^2 + (-1.5)^2 + 0^2}$ $= \sqrt{2.5}$ Exact minimum length is $\sqrt{2.5}$ metres. OR $\overrightarrow{OR} = s\begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$, for some $s \in \mathbb{R}$ $\overrightarrow{PR} = s\begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3s-1 \\ s-2 \\ -2s+1 \end{pmatrix}$ $\left\|\overrightarrow{PR}\right\|^2 = 9s^2 - 6s + 1 + s^2 - 4s + 4 + 4s^2 - 4s + 1$ $= 14s^2 - 14s + 6$ Let $y = PR$. $y^2 = 14s^2 - 14s + 6$ $2y\frac{\mathrm{d}y}{\mathrm{d}s} = 28s - 14$ When $\frac{\mathrm{d}y}{\mathrm{d}s} = 0, 28s - 14 = 0$ $s = \frac{1}{2}$ $2\left(\frac{\mathrm{d}y}{\mathrm{d}s}\right)^2 + 2y\frac{\mathrm{d}^2y}{\mathrm{d}s^2} = 28$ When $\frac{\mathrm{d}y}{\mathrm{d}s} = 0, \quad y > 0$ $\therefore \frac{\mathrm{d}^2y}{\mathrm{d}s^2} = \frac{14}{y} > 0$ *PR* is minimum. The coordinates of *R* are $\left(\frac{3}{2}, \frac{1}{2}, -1\right)$ and exact minimum length is $\sqrt{2.5}$ metres.	If we are not confident of our previous answer, we could also solve this as a shortest distance from *P* to line.

11ia	$\dfrac{dv}{dt}=c$	
b	$v=ct+A$ When $t=0$, $v=4 \Rightarrow A=4$ When $t=2.5$, $v=29 \Rightarrow c=10$ $\therefore v=10t+4$ The value of c is 10.	
ii	$\dfrac{dv}{dt}=10-kv$ $\displaystyle\int \frac{1}{10-kv}\,dv=\int 1\,dt$ Since $10-kv>0$, $-\dfrac{1}{k}\ln(10-kv)=t+C$ $10-kv=Ae^{-kt}$ $v=\dfrac{10-Ae^{-kt}}{k}$ When $t=0$, $v=0$ $0=\dfrac{10-A}{k}\Rightarrow A=10$ $v=\dfrac{10}{k}\left(1-e^{-kt}\right)$	A bit of Physics involved, if $10-kv<0$, the object will fly into the air rather than falling. <u>Alternatively,</u> $\displaystyle\int \frac{1}{10-kv}\,dv=\int 1\,dt$ $-\dfrac{1}{k}\ln\lvert 10-kv\rvert=t+C$ $\lvert 10-kv\rvert=Be^{-kt}$ $\Rightarrow 10-kv=Ae^{-kt}$ The initial velocity for this part is 0, not 4 m s^{-1} from the earlier part.
iii	As $t\to\infty$, $e^{-kt}\to 0$ $40=\dfrac{10}{k}(1-0)$ $k=\dfrac{1}{4}$ $0.9(40)=40\left(1-e^{-\frac{1}{4}t}\right)$ $e^{-\frac{1}{4}t}=0.1$ $t=9.21$ It takes 9.21 s for the object to reach 90% of its terminal velocity.	